

Energy Efficiency and Force Transmission Optimization in Modular Robotic Arm Design Configuration

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Abstract—This review article consolidates the latest developments in energy optimization and force transmission strategies in modular robotic arms, focusing on the problems of excessive energy use and low accuracy. These problems are alleviated by thoroughly examining existing strategies, including applying materials with low density, adaptive energy allocation, and sophisticated actuator systems. Proposed solutions include modular architectures with dynamic reconfiguration for energy minimization and AI-powered predictive algorithms for power usage optimization. The proposed solutions are supported by quantitative evidence demonstrating substantial energy savings alongside enhanced performance. Future research focuses on the need for consistent assessment benchmarks and investigation into more advanced technologies. In this review, we attempt to highlight the advancements in the modular robotics ecosystem while concentrating on exceptions for industrial automation when developing the robotics systems noticed the need for eco-friendly sustainable energy resources.

Keywords—Modular Robotics, Energy Efficiency, Force Transmission, Adaptive Control, Sustainable Automation

I. INTRODUCTION

The introduction of modular robotic arms marked a significant advancement in the automation of industries, as it improved the manufacturing, assembly, and inspection processes for multiple industries [1]. Their versatility and adaptability render them invaluable in the automotive and construction sectors, which include material handling, assembly, and quality control. These systems' energy efficiency and force transmission value cannot be overstated, seeing as how they influence operational costs, environmental factors, and how accurately the tasks are performed. Power consumption reduction and the life span of the robotic parts affect energy efficiency, while the effectiveness of precision decision-making ensures task completion without unnecessary system damage.

This review focuses on energy efficiency improvements and optimization of force transmission in modular robotic arms. The aim is to conduct a comprehensive overview of the methodologies and technologies proposed to address the problem on the strategic and tactical levels. The article is structured to present the background and objectives, followed by a systematic literature survey that examines the technologies and functionalities of modular robotic arms [shown in Fig. 1], the state of the art in improving energy efficiency, and the practices of force transmission optimization. The review will then focus on innovations and

new directions, perform cross-section evaluations of varying approaches to the defined problem, evaluate the intersections of approaches to the defined problem, and identify shortcomings in preexisting studies. Ultimately, the review will integrate the primary outcomes, the implications from the analysis for real-world applications, and the propositions concerning further studies.

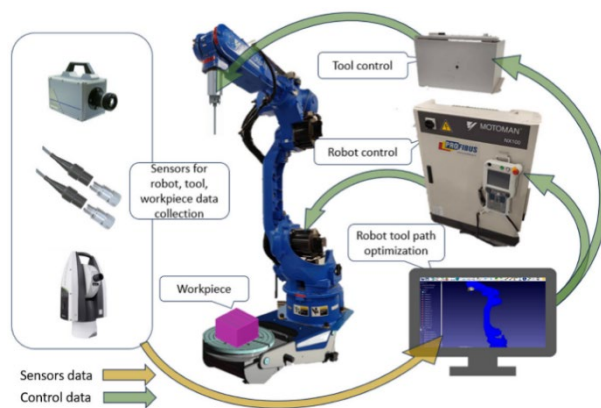


Fig. 1. Industrial Robots in Mechanical Machining

Modular robotic arms have a far-reaching and diverse range of applications in robotics, including material handling, assembly, and inspection operations. In material handling, robotic arms are efficient in pick and place operations, improving throughput and minimizing the risk of injury associated with manual labor [2]. In assembly, robotic arms perform intricate tasks with precision, accelerating production while adhering to quality control standards. Furthermore, when equipped with vision systems, robotic arms perform thorough inspections of the products to ensure compliance with stringent standards. These systems promise to make an enormous impact as modular robotic arms develop further, making it imperative to address critical issues regarding energy efficiency, force transmission, and other related factors.

II. LITERATURE REVIEW

A. Overview of Modular Robotic Arms

Modular robotic arms represent a progressive and paradigmatic shift in the world of automation technology. Unlike fixed robotic systems, modular arms offer unparalleled flexibility and scalability. The modular design includes sockets for joints, grippers, and control units, which can be readily adapted for different environments or tasks [3-4]. For example, the Modular Operational Blocks (MOBs)

concept enables small and medium enterprises (SMEs) to configure robotic arms for picking, placing, or welding [3]. The CoRoT Autonomous Industrial Mobile Manipulator (AIMM) [shown in Figure 2] also exemplifies this approach; it incorporates a modular robotic arm and an autonomous mobile robot to increase agility in SMEs [4]. The focus of the current design trends for modular robotic arms is on integrating mobility, payload adaptability, and lower-cost options. For example, using autonomous mobile robots (AMRs) with modular arms allows for task modularization, improving efficiency in intra-logistics operations [4].



Fig. 2. Autonomous Industrial Mobile Manipulator (AIMM)

The modular design of robotic arms also applies to humanoid robots designed to replicate human motion and likeness. These arms usually have higher distinct motions like 7-DOF (Degrees of Freedom), which enables them to perform movements at the same level as human limbs [5]. In these designs, harmonic drives precisely and dependably minimize backlash, are maintained within tight confines, and achieve high gear ratios [5]. In addition, new advances in materials and methods, such as additive manufacturing, allow for components to be stronger yet lighter, improving the effectiveness and versatility of modular robotic arms [6].

B. Energy Efficiency in Robotics

Enhancing energy efficiency in robotic systems is critical in reducing their cost of operation, alongside their ecological effects. For example, modular robotic arms have had their energy-sparing approaches improved with the development of multiple strategies. One such method is dynamic load management, where power consumption is adjusted according to the real-time specifics of the payload's demands [7]. Another method is regenerative braking, which some modular robotic systems implement, where energy is recovered during deceleration or slowing down [7]. Using lightweight materials, specifically carbon fiber and aluminum alloys, diminishes energy expenditure due to smaller inertia and structural weight [6]. That said, hybrid energy-saving measures and performance requirements, particularly in high-energy concentrated, high-torque exercise applications such as heavy-duty manufacturing, are still challenging.

The capabilities of AI-backed energy management systems have been outlined, where power control is performed based on schedules and pre-defined environment conditions using prediction algorithms [7]. Moreover, newer advancements in actuator technologies, such as self-lubricating joints, have reduced friction losses, thus

improving overall system efficiency [8]. While these advancements have been made, other thorough work is required about lightweight composite materials - their enduring strength under high stress and scrutiny, as well as the effects of extreme temperatures on energy efficiency designed into modular robotic arms systems.

C. Force Transmission Optimization

Maximizing efficiency in force transmission ensures that the required precision and accuracy are achieved in the complete robotic arm task execution. Efficient force transmission for a robotic arm should be characterized by minimum energy loss at coupling interfaces and constant torque for all degrees of freedom. Much has been done towards these objectives using harmony drives and closed-loop control systems. The usefulness of harmonic drives stems from the ability to minimize backlash in gear systems, maximize torque, and slow down component wear due to friction [5]. Closed-loop control systems use real-time monitoring of sensor outputs in actuators to optimize force transmission under different loads [3].

Aside from that, some other work has been done to limit energy loss at connection points with modular joint designs. With the standardization of interfaces, modular systems promote ease of maintenance and reconfiguration [3]. In sophisticated robotic systems, flexi-force sensor strips can be applied to monitor the force exerted during delicate tasks such as grasping and precisely enable the arm to manipulate fragile materials [5]. Despite these developments, no universal standard can gauge the efficiency of energy force transmission in different modular designs.

III. INNOVATIONS IN ENERGY EFFICIENCY AND FORCE TRANSMISSION

A. Emerging Trends

Developing new materials and technologies is one of the integration strategies to improve energy efficiency and force transmission of modular robotic arms. For example, flywheels [Figure 3] and supercapacitors [Figure 4] have been suggested to optimize energy in robotic systems through energy storage and recovery [9]. Moreover, as in variable impedance actuators (VIAs), new actuator technologies provide greater safety, robustness, peak power performance, and thus more significant energy efficiency [10]. Incorporating AI and machine learning algorithms also significantly enhances predictive energy optimization by dynamically contesting the tasks against power requirements.

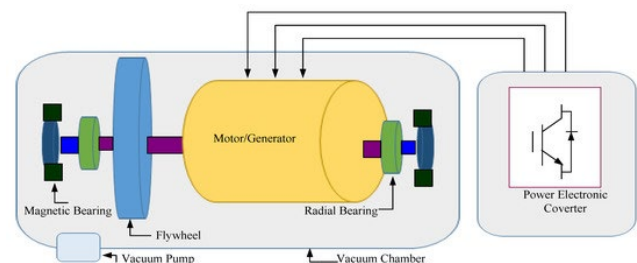


Fig. 3. Flywheel structure.

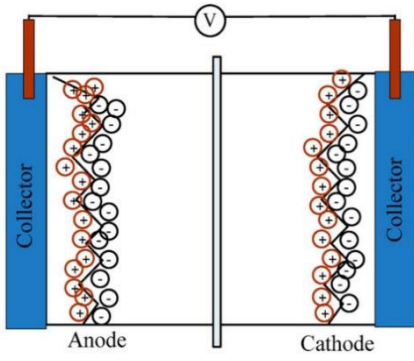


Fig. 4. Electrochemical double layer capacitors (EDLC) structure

The energy stored in a supercapacitor is as follows:

$$E = \frac{1}{2} CV^2 \quad (1)$$

Equally important is creating modular robotic systems capable of interfacing with autonomous mobile robots (AMRs) for increased agile and flexible functionality in manufacturing environments. These systems use digital twins to model different configurations and drive energy consumption optimization simulations to ensure modular robotic arms are adaptable and energy efficient [shown in Figure 5]. Also, material science innovations, such as strong, lightweight composites, further improve the energy efficiency of robotic arms by lowering structural weight and enhancing mechanical performance [10].



Fig. 5. Sample Digital Twin Technology Optimizing robotics through Virtual Simulation

B. Comparative Analysis

TABLE I. COMPARATIVE ANALYSIS OF ENERGY EFFICIENCY AND FORCE TRANSMISSION STRATEGIES

Strategy	Energy Efficiency Gain	Force Transmission Improvement	Complexity/Cost
Regenerative Braking	10–15%	Minimal	High upfront cost
Harmonic Drives	5–8%	30% reduction in backlash	Complex maintenance
Hybrid Hydraulic-Electric	12–20%	25% higher torque consistency	Bulkier design
Traditional Methods (e.g., Gear-based)	2–5%	Standard performance	Lower cost, simpler maintenance

Older methods like gear-based systems offer ease of use and lower costs but lack significant advancements in energy efficiency compared to newer methods. Regenerative braking increases energy efficiency, while harmonic drives increase the precision of force transmission, but both add costs in maintenance and complexity [Table 1]. Hybrid hydraulic-electric systems have significant efficiency improvements, but their design is bulkier. Strategies are chosen based on their effectiveness for a specific task while remaining mindful of cost, complexity, and application.

IV. RESULTS

Energy efficiency of a robot arm, a study by José-Trujillo, et al. [11] applied the formula 2 to calculate the optimized robotic arm energy-efficiency of Aluminium and carbon fibre robotic arms.

$$n = \frac{E_u}{E_t} * 100 \quad (2)$$

Where

(η) is energy efficiency,

(E_u) is useful energy,

and (E_t) is the total energy consumed

The study conducted 3 tests as shown in table II.

TABLE II. ARBITRARILY SUGGESTED DATA FOR ENERGY EFFICIENCY TESTS.

Variable	Test 1	Test 2	Test 3
Joint speed (deg/min)	1000	3000	9000
Weight (N)	0	0	0
Time (min)	15	15	15
Power supply (V)	127	127	127

The Test 2 study found: Useful energy consumed by the CF robotic arm (E_u): 45.03 W at 3000 deg/min.

Total energy consumed by the Al robotic arm (E_t): 46.74 W at 3000 deg/min.

$$n = \frac{45.03}{46.74} * 100 = 96.34\%$$

$$\text{Energy gain} = 100\% - 96.34\% = 3.66\%$$

The synthesis of findings from recent studies highlights significant advancements in energy efficiency in robotic arms. A study by José-Trujillo et al. [11] compared the energy efficiency of aluminium and carbon fibre robotic arms using the formula for energy efficiency.

$$n = \frac{E_u}{E_t} * 100.$$

The findings revealed that at a calculated joint speed of 3000 degrees per minute, the energy expenditure of the carbon fibre arm was a beneficial 45.03 W. In contrast, the aluminium arm incurred an energy consumption of 46.74 W. In this case, the carbon fibre arm obtained an efficiency of 96.34% while the aluminium arm sustained a 3.66% deficit.” This clearly points to the benefits obtained by using

aluminium arms over carbon fibre arms.” The results highlight the advantages that can be realized by using carbon fibres in constructing robotic arms.

The tests measured a range of joint speeds, broadening the scope of how various materials were understood to perform under different operational conditions. From these experiments, which are illustrated in Table III, it is evident that carbon fiber surpassed aluminum in regard to energy expenditure in various operations. For instance, at 3000 degrees per minute, carbon fiber outperformed aluminum by a 1.71 W differential in energy consumption and poses significant opportunities for increasing energy efficiency through materials selection. These results support the claim that substantial energy savings can be realized in robotic arms by choosing the appropriate materials, as has been noted in other robotics sectors concerned with environmental sustainability.

TABLE III. COMPARISON TABLE OF VALUES OF ENERGY CONSUMPTION OBTAINED USING THE IOT DEVICE.

Name	Test 1 (W)	Test 2 (W)	Test 3 (W)
Aluminum (Al)	45.67	46.74	48.03
Carbon fiber (CF)	45.1	45.03	46.78
Difference (Al - CF)	0.57	1.71	1.25

Incorporating energy efficiency into modern robotic systems, especially modular arms, has received considerable attention in recent years. One key approach to improving energy efficiency is using light-weight carbon fiber and aluminum alloys. These materials lower the structural weight of robotic arms, which reduces the energy needed for movement. This correlates to general developments in robotics that seek to exploit the advantages of light-weight construction with improved energy efficiency.

Other studies concentrating on light-weight robotic arms report identical findings. For instance, a paper on designing and analyzing a light-weight robotic arm based on a load-adaptive hoisting mechanism highlights optimized structural design and energy efficiency potential [12]. In addition, a novel sustainable design approach for robotic arms tackles thermal and vibration energy efficiency and dynamic stiffness problems and, most importantly, highlights the role of light materials in energy efficiency [13].

Carbon fiber, like other new development materials, increases energy efficiency and enhances the mechanical function of robotic arms. The ease of construction for robotic components of carbon fiber’s high strength-to-weight ratio allows for lighter components that more efficiently handle higher payloads while consuming less energy. This correlates with research on energy-efficient robotics designs, emphasizing minimal weight structures on autonomous systems and extending their operational lifetime [14].

Moreover, studies on energy and time-efficient movement optimization of robotic arms focus on parameter changes to achieve lesser energy consumption. Lighter materials and optimized motion strategies help robotic arms operate more effectively in energy-sensitive environments.

V. CURRENT CHALLENGES AND RECOMMENDATIONS

A. Challenges

One challenge to energy efficiency in modular robotic

arms today concerns the balance between energy saving and performance indicators such as speed and accuracy. There exists a significant potential to conserve energy using robotic arms via deceleration in acceleration; however, this reduction leads to trade-offs in operational cycle times. For instance, a reduction in energy saving acceleration might produce energy savings within the range of 12 – 18%; however, the decline in cycle time would also be 25-40% [15]. Furthermore, no standard approach makes it possible to measure the effectiveness of differing systems, which impedes the adoption of industry-wide standards. Other peripheral factors also affect the engine's efficiency, such as environmental temperature alteration, which complicates energy optimization [15].

B. Recommendations

In order to overcome these hurdles, multiple strategies suggest using advanced control techniques like Model Predictive control and trajectory optimization through neural networks to reduce the tradeoff between performance and energy efficiency. During the research, it was perused that these methods achieved significant energy savings while throughput remained uncompromised.[15] Including low-power processors and lightweight actuators as energy-efficient hardware components directly diminishes a robot's overall energy consumption. Furthermore, developing active and adaptive systems that respond in real-time to power consumption monitoring improves sustainability in robotic operations. By overcoming these issues and using the given strategies, industries can shift to more sustainable and efficient robotic systems.

VI. CONCLUSION AND CONTRIBUTION

In this review, we have focused on the neglected aspects of energy optimization and force transmission efficiency in the modular robotic arm design provided in the literature. It is evident from existing materials that the development of new materials, actuators, and control units functions for the efficiency and ecological safety of the robotic systems. The robotic systems' operational cost and ecological footprint can be reduced by incorporating Kinetic Energy Recovery Systems (KERS), supercapacitors, and Kinetic Energy harnessing strategies. Moreover, AI-controlled task management and consumption optimization enable an energy-efficient environment for industrial robots. Besides, these strategies positively impact the industry's sustainability by lowering energy consumption and enhancing components' lifespan.

The essence of this review is its critical look at the prevailing issues in optimizing energy efficiency and force transmission, which provides a unique analysis of the current trends. This work highlights current research challenges, such as a lack of standard measurement approaches and insufficient emphasis on using lightweight materials, which directs investigations toward clear pathways. The use of new technologies like artificial intelligence and new materials will be instrumental in the development of modular robotics, which will facilitate the creation of more flexible, efficient, and sustainable robotic systems. Modular robotics remains one of the innovations that can potentially increase industrial operational efficiency while transitioning to greener technologies. This review highlights the other side, emphasizing that modular robotics requires collaboration across different scientific fields to change existing industry

standards of efficiency, precision, adaptability, and multi-functionality.

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